

**Time:** 6 minutes**Outcome:** Calculate trade and capital balances and connect NX, NCO, saving, and investment.**Difficulty:** ●●○ Intermediate

## International Transactions and Open-Economy Identities

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### THE CORE IDEA

Net exports are exports minus imports:  $NX = X - M$ . Positive NX is a trade surplus; negative NX is a trade deficit. Net capital outflow is domestic purchases of foreign assets minus foreign purchases of domestic assets. In the course accounting model,  $NX = NCO$ . National saving finances domestic investment and NCO, so  $S = I + NCO = I + NX$ .

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### HOW TO RECOGNIZE IT

- Exports add to NX; imports subtract from NX.
- A domestic resident buying a foreign asset raises capital outflow and NCO.
- A foreign resident buying a domestic asset lowers NCO and is a capital inflow.
- If S exceeds I, the difference is positive NCO and positive NX.

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### WATCH OUT

Do not import Micro tariff or comparative-advantage analysis into these identities. Also do not call a trade deficit a mathematical error: it corresponds to negative NCO in the accounting model.

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### WORKED EXAMPLE: MATCHING THE TWO SIDES

#### KEY RELATIONSHIPS

- $NX = \text{exports} - \text{imports}$
- $NCO = \text{domestic foreign-asset purchases} - \text{foreign domestic-asset purchases}$
- $NX = NCO$
- $S = I + NCO$
- Therefore  $S = I + NX$

Exports are \$300 billion and imports are \$360 billion, so  $NX = -\$60$  billion. Residents buy \$90 billion of foreign assets while foreigners buy \$150 billion of domestic assets, so  $NCO = \$90 - \$150 = -\$60$  billion. If investment is \$240 billion, saving is  $I + NCO = \$180$  billion.

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### CHECK YOURSELF

National saving is \$250 billion and investment is \$210 billion. What are NCO and NX?

**READY?**

Return to the game and master this concept.